

General and Structural Chemistry
Mid-Semester Examination: Spring 2026
IIIT-Hyderabad

Full Marks: 60

Time: 1:30 Hours

Q.1. (a) What mathematical feature of each of the radial functions controls the number of radial nodes? [1]

(b) Find out the radial node(s) in terms of Bohr radius constants for the ψ_{310} orbital of the hydrogen atom. [3]

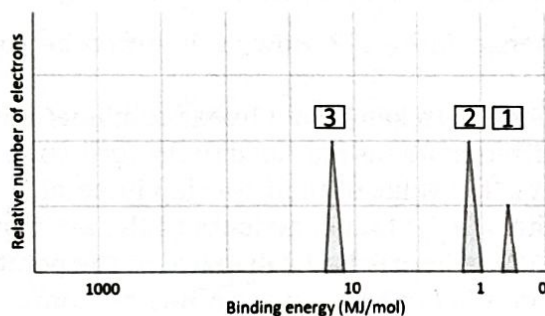
(c) The angular part of the wave function for the d_{xy} orbital is proportional to $\sin^2 \theta \sin 2\phi$. Show that this expression corresponds to the d_{xy} orbital. Also, determine the signs of the lobes. [2,2]

(d) Like $2p_z$ function which is an eigenfunction of $\hat{L}_z$, $2p_x$ and $2p_y$ functions are eigenfunctions of $\hat{L}_x$ and $\hat{L}_y$, respectively. What are their corresponding eigen values? [2]

Q.2. (a) Comment on the angular dependence of the angular wavefunction, $Y_{0,0}$. [2]

(b) For which hydrogen-atom states is ψ nonzero at the nucleus? [1]

(c) Complete photoelectron spectrum of an element in its ground state is presented below. Based on the spectrum, write the ground state electron configuration of the element, and identify the element. Justify your answers. [2,2]



(d) Show that $0 \leq l \leq (n - 1)$, where $n = 1, 2, 3, \dots, \infty$. [3]

Q.3. Answer the following questions about the form of a $4p$ atomic orbital.

(a) How many times does the wavefunction change sign between $r = 0$ and $r = \infty$, keeping θ and ϕ fixed? [2]

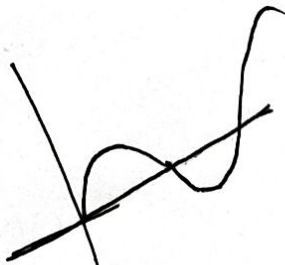
(b) Does the number of angular nodes increase, decrease, or stay the same as the nuclear charge increases? Does the number of radial nodes change? [2,2]

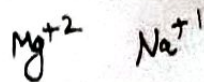
(c) Does the integrated probability density between $r = 0$ and $r = a_0$ increase, decrease, or stay the same as the nuclear charge increases? [2]

(d) What is the numerical value for the orbital angular momentum of an electron in this orbital? [2]

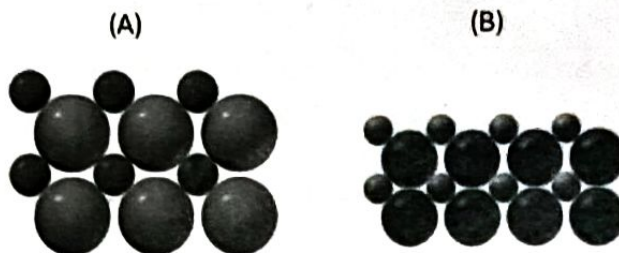
Q.4. (a) Although $3d$ orbital has lower energy than $4s$, we have seen that in iron ($Z = 26$), the electron configuration $4s^2 3d^6$ is favored over $3d^8 4s^0$. Discuss why the former electron configuration is favored for iron. [4]

4-1-1=2





(b) One of the following pictures represents MgO and one represents NaCl. Which is which? Give reasoning. [3]



(c) Compare the ionization energies of the atomic hydrogen (H) and molecular hydrogen (H₂). Justify your answer. [2]

(d) The series 2 elements are more likely to form multiple bonds which are almost nonexistent in their heavier group members. Explain. [1]

Q.5. (a) Rationalize the following observations based on the electronic structures of the atoms and their ions. "Hydrogen chloride may be prepared from sodium chloride and sulfuric acid, but the same method cannot be employed in the preparation of hydrogen bromide or hydrogen iodide. [4]

(b) Give an example of an auride compound. Point out all the possible factors that lead to the auride formation. [1,2]

(c) Across 3rd period, the expected trend of the change in diameter sees a conflict between that phosphorus and sulfur. Discuss the above observation. [3]

Q.6. (a) What is 'exchange integral'? How is it different from the 'overlap integral'? [1,1]

(b) Draw schematic diagrams to show how linear combinations of an s-orbital and a p-orbital on two different atoms may lead to the formation of a nonbonding molecular orbital. What is the value of their overlap integral. [1,1]

(c) The two atomic orbitals (ϕ_i 's) that contribute to the antibonding orbital are each proportional to e^{-r_i/a_0} , where r_i is the distance of the point from its parent nucleus i (say, from atom A is r_A and from atom B is r_B). Confirm that there is a nodal plane lying halfway between the two nuclei. [2]

(d) Write down the correct ground state molecular wave functions (including spin function) for H₂ molecule according to LCAO-MO and VB methods (including the ionic contributions). [2, 2]

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GIVEN: The terms bear their usual significance.

$$\psi_{310} = \frac{\sqrt{2}}{81\sqrt{\pi}a_0^{3/2}} \left(6 - \frac{r}{a_0}\right) \frac{r}{a_0} e^{-r/3a_0} \cos \theta \quad x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sin(a + b) = \sin a \cos b + \cos a \sin b \quad \cos(a + b) = \cos a \cos b - \sin a \sin b$$

$$R_{n,l}(r) = -\sqrt{\left(\frac{2Z}{na_0}\right)^3 \frac{(n-l-1)!}{2n[(n+l)!]^3}} \cdot \left(\frac{2Zr}{na_0}\right)^l \cdot e^{-Zr/na_0} \cdot L_{n+l}^{2l+1}\left(\frac{2Zr}{na_0}\right)$$

$$2l+1 > n-l$$

$$2l+1$$

$$n+l < 2l+1$$

$$n-$$